Report

Clinical Application – Retinal Vessel Segmentation

Chapter 1: Problem Definition

This project explores the clinical use of mathematical morphology in digital image processing for isolating retinal blood vessels from color fundus images. The retinal vasculature provides critical diagnostic cues for systemic and ocular diseases such as diabetic retinopathy, hypertension, and glaucoma. Accurate vessel segmentation enables precise computation of vascular features such as **caliber**, **tortuosity**, and **arteriolar–venular ratio**, and serves as a vital preprocessing step in machine learning-based disease classification pipelines.

The goal of this task is to **generate a high-contrast binary vessel mask** using exclusively **spatial morphological operations**. In the final binary mask:

- Pixels labeled **1 (white)** correspond strictly to blood vessels.
- Pixels labeled **0 (black)** correspond to all other retinal structures.

Such a binary map is highly valuable for clinical diagnostics and quantitative biomedical analysis.

Chapter 2: Solution Methodology and Implementation Details

The pipeline adheres strictly to morphological image processing and manual histogram-based analysis. No high-level libraries or built-in morphology functions were used beyond basic I/O (cv2.imread, matplotlib, etc.). All core operations were coded from scratch to align with assignment constraints.

1. Channel Selection

- The color fundus image was loaded from '/content/sample-image.jpg'.
- Only the **green channel** was extracted:

  ```
  green = img[:, :, 1].astype(np.uint8)
  ```

- Rationale: The green channel offers **maximum vessel-to-background contrast**, making it optimal for vascular segmentation.

2. Vessel Enhancement

A robust vessel enhancement process was applied in multiple stages:

a. Background Estimation via Morphological Opening

- A large disk-shaped structuring element (radius = 28) was used for gray-scale **opening**:

  se_bg = disk(28)
  background = gray_opening(green, se_bg)

- This removes small bright details (like vessels), retaining the slow-varying background.

b. Flat-Field Correction

- Subtracting the estimated background from the original green channel:

  flat = green - background

- This normalizes illumination and improves uniformity across the field.

c. Manual Contrast Normalization

- The flat-fielded image was stretched manually to span [0, 255] using a custom normalization function:

  flat_norm = normalize_manual(flat)

d. Multi-Scale White Top-Hat on Inverted Image

- The normalized image was inverted to make vessels bright:

  inv_flat = 255 - flat_norm

- A **white top-hat transform** (original − opening) was applied with 3 different structuring element sizes to enhance vessels of varying width:

```
enh_s1 = white_tophat(inv_flat, disk(5))  # fine vessels
enh_s2 = white_tophat(inv_flat, disk(7))  # small vessels
enh_s3 = white_tophat(inv_flat, disk(9))  # medium vessels
```

- These were combined using:

```
enhanced = np.maximum.reduce([enh_s1, enh_s2, enh_s3])
```

- This approach captures both thin capillaries and larger trunk vessels more effectively than single-scale filtering.

3. Binary Segmentation

a. Otsu's Thresholding (Implemented from Scratch)

- A custom function computed the global threshold by maximizing between-class variance:

```
T = otsu_threshold(enhanced)
```

b. Binarization

- The enhanced image was thresholded to form a binary vessel mask:

```
mask_otsu = (enhanced >= T).astype(np.uint8) * 255
```

4. Morphological Clean-up and Reconstruction

To refine the mask and improve vessel connectivity:

a. Marker-Based Binary Reconstruction (Hysteresis)

- A marker image was created by eroding the binary mask with disk(3) to retain vessel centerlines:

```
marker = bin_erosion(mask_otsu, disk(3))
recon = binary_reconstruct(marker, mask_otsu)
```

b. Morphological Tidy-Up

- To seal small gaps and remove noise:

```
recon = bin_closing(recon, disk(5))
recon = bin_opening(recon, se3)  # 3×3 kernel
```

c. Area Opening (Custom Implementation)

- A **BFS-based area opening function** was implemented to discard small disconnected regions:

  ```
  recon = area_opening(recon, min_area=100)
  ```

d. Retina ROI Masking

- A custom manual threshold was used to create a retina field-of-view mask:

  ```
  retina_roi = manual_threshold(green, thresh=15)
  retina_roi = bin_closing(retina_roi, disk(10))
  ```

- All pixels outside the retinal field were masked out:

  ```
  recon[~retina_roi] = 0
  ```

Final Output:

```
mask_clean = recon.copy()
```

Chapter 3: Results and Discussion

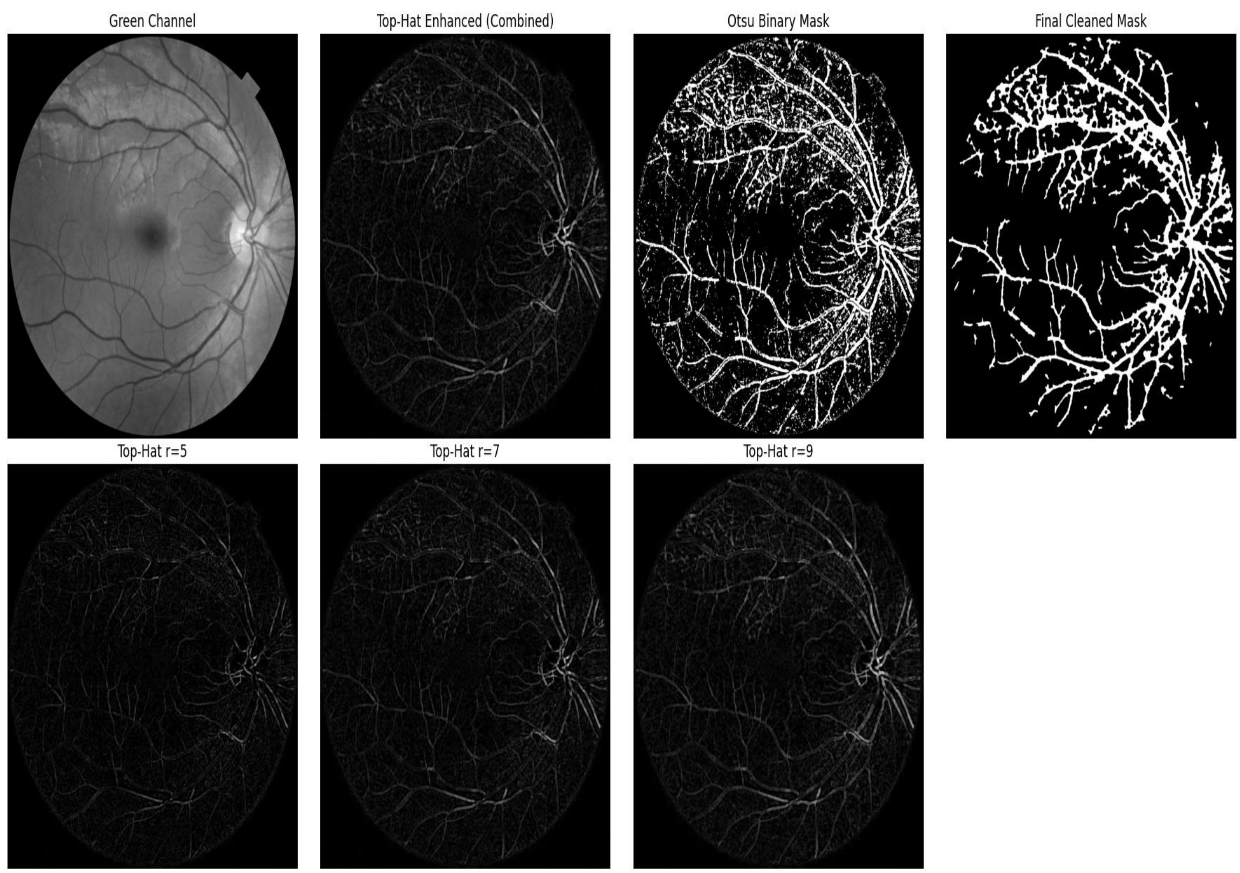
The full processing pipeline was visualized in two rows:

- **Row 1**: Major processing stages
  - Green Channel
  - Merged Enhanced Image
  - Otsu Thresholded Mask
  - Final Cleaned Binary Mask
- **Row 2**: Top-Hat Enhancements at different scales
  - Top-Hat with radius 5
  - Top-Hat with radius 7
  - Top-Hat with radius 9

Key Observations:

- **Contrast Stretching** helped significantly in improving vessel separation.
- **Multi-scale Top-Hat** outperformed single-scale filtering by preserving both fine and coarse vessel branches.
- The **custom binary reconstruction** successfully reconnected fragmented vessels, demonstrating the power of hysteresis-like morphological filling.
- **Hand-coded area opening** eliminated false positives caused by noise or isolated pixel clusters, ensuring higher specificity.
- Masking outside the retina field refined the output and made it clinically realistic.

Figure 1: Intermediate Vessel Enhancement Stages



Conclusion

This project successfully demonstrated how spatial morphological operations—when carefully sequenced and implemented from scratch—can produce a reliable binary vessel mask from a retinal fundus image. The multi-scale enhancement strategy, custom Otsu thresholding, and morphology-based reconstruction yielded high-quality segmentation results, ready for downstream feature extraction or clinical analysis.